

Spinor Transfer Model from J3 to J2 Space in Collatz Inverse Trees

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Abstract

This report presents a theoretical model that reconstructs the generation mechanism of Collatz Inverse Trees as a "Spinor Transfer" from the 3-adic Jacobsthal (J3) space to the 2-adic Jacobsthal (J2) space. While inverse tree generation relies on the reverse calculation of $3n + 1$, this study demonstrates a structure capable of generating the inverse tree using only addition and phase (mod 3), completely eliminating the need for multiplication and division. For an odd non-multiple of 3, denoted as b , its local phase (Balanced Ternary) in the J3 space uniquely determines the J3 spinor (a, b) . This spinor is then linearly transferred to the J2 spinor $(b - a, b + a)$ in the J2 space. Subsequently, branches radiate according to the J2 recurrence relation:

$$J_3(a, b) \rightarrow J_2(b - a, b + a)$$

$$x_{k+2} = x_{k+1} + 2x_k$$

Through this transfer model, we show that the branching structure of the Collatz inverse tree can be naturally described as the geometric interlocking (resonance) of two adic spaces.

1. Definition of Fundamental Sequences and Spatial Expansion

In this study, the odd nuclei in the Collatz map are represented by the following two Jacobsthal systems:

- **3-adic Jacobsthal (J3) Space:** Determines the local phase.
- **2-adic Jacobsthal (J2) Space:** Governs branch extension (exponential expansion).

1.1. 2-adic Jacobsthal (J2) Space

The J2 space governs the "binary extension" of the Collatz inverse tree.

$$J_{2,k+2} = J_{2,k+1} + 2J_{2,k}, \quad J_{2,0} = 0, \quad J_{2,1} = 1$$

Example of J2 expansion:

$$0, 1, 1, 3, 5, 11, 21, 43, \dots$$

1.2. 3-adic Jacobsthal (J3) Space

The J3 space governs the "ternary local phase" of the odd nuclei. In this study, we adopt the following linear recurrence relation as the Extended Jacobsthal System.

$$J_{3,k+2} = 2J_{3,k+1} + 3J_{3,k}, \quad J_{3,0} = 0, \quad J_{3,1} = 1$$

Example of J3 expansion:

$$0, 1, 2, 7, 20, 61, 182, \dots$$

2. J3 Spinor and Balanced Ternary Phase

For an odd nucleus n , the lowest 3-adic phase (Balanced Ternary) is defined as follows:

$$\text{Phase}(n) \in \{-1, 0, +1\}$$

$$\text{Phase}(n) = \begin{cases} 0 & (n \equiv 0 \pmod{3}) \Rightarrow a = \frac{n}{3} \\ +1 & (n \equiv 1 \pmod{3}) \Rightarrow a = \frac{n-1}{3} \\ -1 & (n \equiv 2 \equiv -1 \pmod{3}) \Rightarrow a = \frac{n+1}{3} \end{cases}$$

Let the current odd nucleus be $b = n$. The 2-component vector (a, b) is called the J3 Spinor (referring to a linear 2-component vector in this paper).

3. Spinor Transfer Model: J3 to J2

3.1. Transfer Equation (Linear Mapping)

The 3-adic information of the J3 spinor is transferred to the J2 spinor via the following linear mapping:

$$J_3(a, b) \rightarrow J_2(b - a, b + a)$$

This is a linear transformation where the local phase of J3 is directly reflected in the initial conditions of J2.

$$T(a, b) = (b - a, b + a)$$

The transferred J2 spinor radiates into space according to the J2 recurrence relation:

$$(x_1, x_2) = (b - a, b + a)$$

$$x_{k+2} = x_{k+1} + 2x_k$$

3.2. Interlocking of Balanced Ternary Phase and J2 Parity

The even/odd parity of the J2 sequence naturally classifies the parent nodes of the following Collatz maps:

$$3n + 1, \quad 3n - 1$$

- Phase = +1
 - Even terms $x_{2m} \rightarrow$ Parent in the $3n + 1$ world.
 - Odd terms $x_{2m-1} \rightarrow$ Parent in the $3n - 1$ world.
- Phase = -1
 - Odd terms $x_{2m-1} \rightarrow$ Parent in the $3n + 1$ world.
 - Even terms $x_{2m} \rightarrow$ Parent in the $3n - 1$ world.

Through this structure, the parent nodes of the Collatz inverse tree are naturally extracted as the even/odd parity of the J2 sequence.

4. Verification Example

In this section, we demonstrate that the model perfectly aligns with the actual Collatz inverse tree for the initial value 7: $1 \leftarrow 5 \leftarrow 13 \leftarrow 17 \leftarrow 11 \leftarrow 7$

4.1. $n = 1$ (*Phase* = +1)

- J3 Spinor: $J_3(0,1)$
- J2 Transfer: $J_2(1,1)$
- J2 Expansion: 1, 1, 3, **5**, 11, ...
- Parent Node: The even term 5 is selected. ($3 \times 5 + 1 = 16 \rightarrow 1$)

4.2. $n = 5$ (*Phase* = -1)

- J3 Spinor: $J_3(2,5)$
- J2 Transfer: $J_2(3,7)$
- J2 Expansion: 3, 7, **13**, 27, ...
- Parent Node: The odd term 13 is selected. ($3 \times 13 + 1 = 40 \rightarrow 5$)

4.3. $n = 13$ (*Phase* = +1)

- J3 Spinor: $J_3(4,13)$
- J2 Transfer: $J_2(9,17)$
- J2 Expansion: 9, **17**, 35, 69, ...
- Parent Node: The even term 17 is selected. ($3 \times 17 + 1 = 52 \rightarrow 13$)

4.4. $n = 17$ (*Phase* = -1)

- J3 Spinor: $J_3(6,17)$
- J2 Transfer: $J_2(11,23)$
- J2 Expansion: **11**, 23, 45, 91, ...
- Parent Node: The odd term 11 is selected. ($3 \times 11 + 1 = 34 \rightarrow 17$)

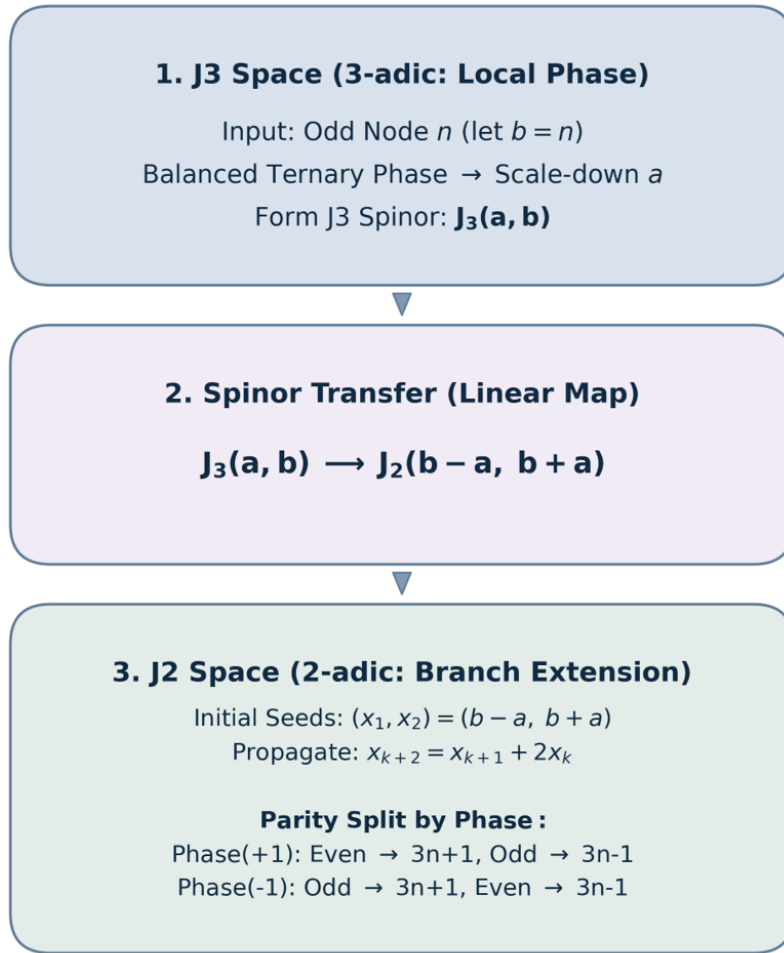
4.5. $n = 11$ (*Phase* = -1)

- J3 Spinor: $J_3(4,11)$
- J2 Transfer: $J_2(7,15)$
- J2 Expansion: **7**, 15, 29, 59, ...
- Parent Node: The odd term 7 is selected. ($3 \times 7 + 1 = 28 \rightarrow 11$)

Thus, the Collatz inverse tree for the initial value 7 completely matches the sequence of J3 $\rightarrow$ J2 spinor transfers (**Figure 1**).

(a)

Conceptual Architecture of Spinor Transfer ($J_3 \rightarrow J_2$)



(b)

Node ($n = b$)	Phase	$J_3(a, b)$	$J_2(b-a, b+a)$	J2_Sequence
7	1	$J_3(2, 7)$	$J_2(5, 9)$	[5, 9, 19, 37, 75]
11	-1	$J_3(4, 11)$	$J_2(7, 15)$	[7, 15, 29, 59, 117]
17	-1	$J_3(6, 17)$	$J_2(11, 23)$	[11, 23, 45, 91, 181]
13	1	$J_3(4, 13)$	$J_2(9, 17)$	[9, 17, 35, 69, 139]
5	-1	$J_3(2, 5)$	$J_2(3, 7)$	[3, 7, 13, 27, 53]
1	1	$J_3(0, 1)$	$J_2(1, 1)$	[1, 1, 3, 5, 11]

Figure 1. Spinor Transfer Architecture and Computed Orbit.

(a) Schematic representation of the linear spinor transfer mapping from the 3-adic (J_3) local phase to the 2-adic (J_2) branch-extension space. (b) Spinor values computed along the reverse odd Collatz orbit for the initial value 7 ($1 \leftarrow 5 \leftarrow 13 \leftarrow 17 \leftarrow 11 \leftarrow 7$), highlighting the parity-based parent selection governed by the Balanced Ternary phase.

5. Conclusion

In this study, the Collatz inverse tree was formalized as the following linear spinor transfer model:

$$J_3(a, b) \rightarrow J_2(b - a, b + a)$$

This clarified a geometric and deterministic mechanism for inverse tree generation:

- Local phase in the 3-adic world (Balanced Ternary).
- Linear transfer to the initial conditions of exponential expansion in the 2-adic world.
- Radiation of inverse tree branches via the J2 recurrence relation.

Transcending the procedural understanding of "reverse calculating $3n + 1$," it is shown that the Collatz inverse tree is naturally generated as the resonance structure of J3 and J2 adic spaces.

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